

**Tables —: Schemes for internal assessments and end semester
(Pharmacology-MPL)**

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks	Credit Score	
		Continuous Mode	Sessional Exams		Total	Marks	Duration		Credit Hours	Credit Points
			Marks	Duration						
SEMESTER III										
MRM301T	Research Methodology & Biostatistics*	10	15	1 Hr	25	75	3 Hrs	100	4	4
-	Journal club	-	-	-	25	-	-	25	1	1
-	Discussion / Presentation (Proposal Presentation)	-	-	-	50	-	-	50	2	2
MPL302(P)	Research work*	-	-	-	-	350	1 Hr	350	28	14
TOTAL								525	35	21

*Non University Examination

BHABHA UNIVERSITY



Syllabus

**M.Pharm III Sem
(Pharmacology)**

Semester III

MRM301T - Research Methodology & Biostatistics

UNIT – I

General Research Methodology: Research, objective, requirements, practical difficulties, review of literature, study design, types of studies, strategies to eliminate errors/bias, controls, randomization, crossover design, placebo, blinding techniques.

UNIT – II

Biostatistics: Definition, application, sample size, importance of sample size, factors influencing sample size, dropouts, statistical tests of significance, type of significance tests, parametric tests (students “t” test, ANOVA, Correlation coefficient, regression), non-parametric tests (wilcoxon rank tests, analysis of variance, correlation, chi square test), null hypothesis, P values, degree of freedom, interpretation of P values.

UNIT – III

Medical Research: History, values in medical ethics, autonomy, beneficence, non-maleficence, double effect, conflicts between autonomy and beneficence/non-maleficence, euthanasia, informed consent, confidentiality, criticisms of orthodox medical ethics, importance of communication, control resolution, guidelines, ethics committees, cultural concerns, truth telling, online business practices, conflicts of interest, referral, vendor relationships, treatment of family members, sexual relationships, fatality.

UNIT – IV

CPCSEA guidelines for laboratory animal facility: Goals, veterinary care, quarantine, surveillance, diagnosis, treatment and control of disease, personal hygiene, location of animal facilities to laboratories, anesthesia, euthanasia, physical facilities, environment, animal husbandry, record keeping, SOPs, personnel and training, transport of lab animals.

UNIT – V

Declaration of Helsinki: History, introduction, basic principles for all medical research, and additional principles for medical research combined with medical care.